Chapter 1.3 Random Variables

Rick Durrett Probability: Theory and Examples

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Built from the local Chapter 1 LLMviki flash cards.

Roadmap for 1.3

Learning goal

Prove measurability quickly using generators and closure under limits/composition.

1. Random variables as measurable maps.
2. Generator checking and inverse images.
3. Arithmetic, composition, and limits of random variables.
4. Semicontinuity and discontinuity sets.
5. $\sigma(X)$ factorization: when $Y = f(X)$.

Random Variables as Measurable Maps

Definition

A random variable is a measurable map

$$X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R})).$$

Equivalently,

$$\{X \in B\} \in \mathcal{F} \qquad \text{for every } B \in \mathcal{B}(\mathbb{R}).$$

Pitfall

A random variable is not the same thing as its distribution. Different maps may have the same law.

LLMviki: `Random variables as measurable maps; ID: durrett_ch1_measurable_map`

Theorem

If $\mathcal{S} = \sigma(\mathcal{A})$, then to prove $X : (\Omega, \mathcal{F}) \to (E, \mathcal{S})$ is measurable, it is enough to check

$$X^{-1}(A) \in \mathcal{F} \qquad \text{for every } A \in \mathcal{A}.$$

Let

$$\mathcal{D} = \{B \in \mathcal{S} : X^{-1}(B) \in \mathcal{F}\}.$$

Since inverse images preserve complements and countable unions, $\mathcal{D}$ is a sigma-field. If $\mathcal{A} \subseteq \mathcal{D}$, then $\sigma(\mathcal{A}) \subseteq \mathcal{D}$, so all target measurable sets have measurable inverse image.

LLMviki: `Checking measurability on generators; ID: durrett_ch1_measurability_generator`

Composition of Measurable Maps

Theorem

If X is measurable and f is measurable, then $f(X)$ is measurable.

For any Borel set B,

$$\{f(X) \in B\} = \{X \in f^{-1}(B)\}.$$

Since f is measurable, $f^{-1}(B)$ is Borel; since X is measurable, the event on the right is in $\mathcal{F}$.

LLMviki: `Composition of measurable maps; ID: durrett_ch1_composition_measurable`

Arithmetic of Random Variables

Theorem

Finite sums, products, maxima, minima, and continuous functions of random variables are random variables.

- Vector trick: first prove (X, Y) is measurable.
- Then compose with continuous maps such as $(x, y) \mapsto x + y$ or $(x, y) \mapsto xy$.
- Direct trick for sums: use rational cuts.

LLMviki: `Arithmetic of random variables; ID: durrett_ch1_measurable_arithmetic`

Limits Preserve Measurability

Theorem

If (X_n) are measurable real functions, then

$$\sup_n X_n, \quad \inf_n X_n, \quad \limsup_n X_n, \quad \liminf_n X_n$$

are measurable. Any pointwise limit of measurable functions is measurable.

Use countable descriptions such as

$$\left\{\sup_n X_n \le a\right\} = \bigcap_{n=1}^{\infty} \{X_n \le a\}.$$

Then write limsup/liminf through countable suprema and infima:

$$\limsup_n X_n = \inf_m \sup_{n \ge m} X_n.$$

LLMviki: `Limits preserve measurability; ID: durrett_ch1_limits_measurable`

$\sigma(X)$ Factorization

Fact

A random variable Y is $\sigma(X)$-measurable if and only if there is a Borel measurable $f : \mathbb{R} \to \mathbb{R}$ such that

$$Y = f(X).$$

- Easy direction: inverse images compose.
- Hard direction: approximate Y by simple functions or dyadic bins.

LLMviki: `Functions measurable with respect to sigma(X); ID: durrett_ch1_sigma_x_factorization`

Exercise 1.3.1

Show that if $\mathcal{A}$ generates $\mathcal{S}$, then

$$X^{-1}(\mathcal{A}) \equiv \{\{X \in A\} : A \in \mathcal{A}\}$$

generates

$$\sigma(X) = \{\{X \in B\} : B \in \mathcal{S}\}.$$

LLMviki: `Checking measurability on generators; ID: durrett_ch1_measurability_generator`

Exercise 1.3.1: Proof Skeleton

Let $\mathcal{G} = \sigma(X^{-1}(\mathcal{A}))$. Since $\mathcal{A} \subseteq \mathcal{S}$, $\mathcal{G} \subseteq \sigma(X)$. For the reverse inclusion, define

$$\mathcal{D} = \{B \in \mathcal{S} : \{X \in B\} \in \mathcal{G}\}.$$

The class $\mathcal{D}$ is a sigma-field and contains $\mathcal{A}$, hence $\mathcal{D} = \mathcal{S}$. Therefore every $\{X \in B\}$ with $B \in \mathcal{S}$ lies in $\mathcal{G}$.

LLMviki: `Checking measurability on generators; ID: durrett_ch1_measurability_generator`

Exercise 1.3.2

Prove Theorem 1.3.6 when $n = 2$ by checking directly that

$$\{X_1 + X_2 < x\} \in \mathcal{F}.$$

LLMviki: `Rational cut proof for sum measurability; ID: exercise_method_rational_cut_sum_measurability`

Exercise 1.3.2: Proof Skeleton

For fixed $x \in \mathbb{R}$, write

$$\{X_1 + X_2 < x\} = \bigcup_{q \in \mathbb{Q}} \left(\{X_1 < q\} \cap \{X_2 < x - q\}\right).$$

The equality follows from density of $\mathbb{Q}$ between $X_1(\omega)$ and $x - X_2(\omega)$. The right side is a countable union of measurable events.

LLMviki: `Rational cut proof for sum measurability; ID: exercise_method_rational_cut_sum_measurability`

Exercise 1.3.3

Show that if f is continuous and $X_n \to X$ almost surely, then

$$f(X_n) \to f(X)$$

almost surely.

LLMviki: `Composition of measurable maps; ID: durrett_ch1_composition_measurable`

Exercise 1.3.3: Proof Skeleton

Let

$$\Omega_0 = \{\omega : X_n(\omega) \to X(\omega)\}.$$

Then $\mathbb{P}(\Omega_0) = 1$. For $\omega \in \Omega_0$, continuity of f gives

$$f(X_n(\omega)) \to f(X(\omega)).$$

Hence convergence holds on a probability-one set.

LLMviki: `Composition of measurable maps; ID: durrett_ch1_composition_measurable`

Exercise 1.3.4

(i) Show that a continuous function from $\mathbb{R}^d$ to $\mathbb{R}$ is measurable from $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

(ii) Show that $\mathcal{B}(\mathbb{R}^d)$ is the smallest sigma-field that makes all continuous functions measurable.

LLMviki: `Composition of measurable maps; ID: durrett_ch1_composition_measurable`

Exercise 1.3.4: Proof Skeleton

For (i), inverse images of open sets under continuous functions are open, hence Borel. For (ii), suppose $\mathcal{G}$ makes every continuous real function measurable. For any closed $C \subseteq \mathbb{R}^d$, the distance function

$$d_C(x) = \inf_{y \in C} |x - y|$$

is continuous and

$$C = \{x : d_C(x) = 0\}.$$

Thus all closed sets, hence all Borel sets, lie in $\mathcal{G}$.

LLMviki: `Checking measurability on generators; ID: durrett_ch1_measurability_generator`

Exercise 1.3.5

A function f is lower semicontinuous if

$$\liminf_{y \to x} f(y) \geq f(x),$$

and upper semicontinuous if $-f$ is lower semicontinuous. Show that f is lower semicontinuous if and only if $\{x : f(x) \leq a\}$ is closed for each $a \in \mathbb{R}$, and conclude that semicontinuous functions are measurable.

LLMviki: `Lower semicontinuity via closed sublevel sets; ID: exercise_method_closed_sublevel_semicontinuity`

Exercise 1.3.5: Proof Skeleton

If f is l.s.c. and $x_n \to x$ with $f(x_n) \le a$, then

$$f(x) \le \liminf_n f(x_n) \le a,$$

so $\{f \le a\}$ is closed. Conversely, if all $\{f \le a\}$ are closed and $\liminf f(y_n) < f(x)$ for some $y_n \to x$, choose a between them and get a contradiction by closedness. Since $\{f \le a\}$ is Borel for every a, f is measurable.

LLMviki: `Lower semicontinuity via closed sublevel sets; ID: exercise_method_closed_sublevel_semicontinuity`

Exercise 1.3.6

Let $f : \mathbb{R}^d \to \mathbb{R}$ be arbitrary and define

$$f^\delta(x) = \sup\{f(y) : |y - x| < \delta\}, \qquad f_\delta(x) = \inf\{f(y) : |y - x| < \delta\}.$$

Show that f^δ is lower semicontinuous and f_δ is upper semicontinuous. Let

$$f^0 = \lim_{\delta \downarrow 0} f^\delta, \qquad f_0 = \lim_{\delta \downarrow 0} f_\delta.$$

Conclude that the set of discontinuities of f is $\{f^0 \neq f_0\}$ and is measurable.

LLMviki: `Discontinuity set from upper and lower local envelopes; ID: exercise_method_oscillation_discontinuity_set`

Exercise 1.3.6: Proof Skeleton

Show $\{x : f^\delta(x) > a\}$ is open: if $f^\delta(x) > a$, some nearby y has $f(y) > a$, and the same y remains nearby for all x' close to x. Thus f^δ is l.s.c.; similarly f_δ is u.s.c. The limits f^0, f_0 are measurable. Continuity at x is equivalent to the small-ball upper and lower envelopes squeezing together:

$$f^0(x) = f_0(x).$$

LLMviki: `Discontinuity set from upper and lower local envelopes; ID: exercise_method_oscillation_discontinuity_set`

Exercise 1.3.7

A function $\phi : \Omega \to \mathbb{R}$ is simple if

$$\phi(\omega) = \sum_{m=1}^{n} c_m \mathbf{1}_{A_m}(\omega), \qquad A_m \in \mathcal{F}.$$

Show that the class of $\mathcal{F}$-measurable functions is the smallest class containing the simple functions and closed under pointwise limits.

LLMviki: `Limits preserve measurability; ID: durrett_ch1_limits_measurable`

Exercise 1.3.7: Proof Skeleton

Simple functions are measurable, and pointwise limits of measurable functions are measurable. For minimality, approximate any nonnegative measurable f by simple functions

$$f_n = 2^{-n} \lfloor 2^n (f \wedge n) \rfloor .$$

Then $f_n \to f$ pointwise. For general real f, apply this to f^+ and f^- and take differences.

LLMviki: `Limits preserve measurability; ID: durrett_ch1_limits_measurable`

Exercise 1.3.8

Use the previous exercise to conclude that Y is measurable with respect to $\sigma(X)$ if and only if

$$Y = f(X)$$

for some measurable function $f : \mathbb{R} \to \mathbb{R}$.

LLMviki: `Functions measurable with respect to sigma(X); ID: durrett_ch1_sigma_x_factorization`

Exercise 1.3.8: Proof Skeleton

If $Y = f(X)$, then $\{Y \in B\} = \{X \in f^{-1}(B)\} \in \sigma(X)$. Conversely, let $\mathcal{H}$ be the class of $\sigma(X)$-measurable functions representable as $f(X)$. It contains simple $\sigma(X)$-measurable functions and is closed under pointwise limits. By Exercise 1.3.7, every $\sigma(X)$-measurable Y belongs to $\mathcal{H}$.

LLMviki: `Functions measurable with respect to sigma(X); ID: durrett_ch1_sigma_x_factorization`

Exercise 1.3.9

Give a constructive proof of the previous result. Note that

$$\{\omega : m2^{-n} \leq Y(\omega) < (m+1)2^{-n}\} = \{X \in B_{m,n}\}$$

for some Borel set $B_{m,n}$, and define $f_n(x) = m2^{-n}$ for $x \in B_{m,n}$. Show that $f_n(x) \to f(x)$ and $Y = f(X)$.

LLMviki: `Constructive representation of sigma(X)-measurable functions; ID: exercise_method_constructive_sigma_x_representation`

Exercise 1.3.9: Proof Skeleton

Partition Ω into dyadic bins

$$A_{m,n} = \{m2^{-n} \le Y < (m+1)2^{-n}\}.$$

Since $A_{m,n} \in \sigma(X)$, write $A_{m,n} = \{X \in B_{m,n}\}$. Make the $B_{m,n}$ disjoint if needed, then define

$$f_n(x) = \sum_{m \in \mathbb{Z}} m2^{-n}\mathbf{1}_{B_{m,n}}(x).$$

Then $0 \le Y - f_n(X) < 2^{-n}$, so $f_n(X) \to Y$. Let $f = \lim f_n$ on the Borel set where the limit exists, and 0 elsewhere.

LLMviki: `Constructive representation of sigma(X)-measurable functions; ID: exercise_method_constructive_sigma_x_representation`

Checklist: How to Attack 1.3 Problems

1. To prove measurability, reduce to a generator.
2. For sums, use rational cuts when a direct proof is requested.
3. For limits, rewrite events using countable unions/intersections.
4. For semicontinuity, use closed sublevel sets or open superlevel sets.
5. For $\sigma(X)$-measurability, think "information carried by X" and prove $Y = f(X)$.

LLMviki: `Chapter 1.3 flash cards and exercise method cards`

Mini Practice

1. Prove that $\max(X, Y)$ is measurable using rational cuts.
2. If X_n are random variables, show $\{\lim_n X_n \text{ exists}\} \in \mathcal{F}$.
3. Show every continuous $f : \mathbb{R}^d \to \mathbb{R}^k$ is Borel measurable.
4. If Y is $\sigma(X)$-measurable and g is Borel, explain why $g(Y)$ is $\sigma(X)$-measurable.